

Machine Learning Based Multi-Criteria Evaluation of Six Developed Countries for Mobility Decision Support

Abstract—An intelligent machine learning-based system that makes country recommendations is developed to assist people who are relocating internationally for education and/or employment. In this paper, six developed countries, including the United States, Canada, UK, Germany, Australia and Japan, will be evaluated against a set of seven socio-economic indicators related to quality of life, safety, healthcare, cost of living, pollution, purchasing power and education. The numerical indicators associated with each of these seven socio-economic indicators have been transformed into order categorical features using domain specific feature engineering and ordinal encoding. With the LazyPredict framework, several supervised learning models were evaluated. The model that performed the best with good stability across all tasks was Gaussian Naïve Bayes due to the limited amount of training data available. The system incorporates user personalization, a visual representation of the results, and an example of the LIME based explainability method, all of which make it easier to interpret the recommendations. The recommendations results are accurate, robust, and easily understood in relation to the decision making associated with international mobility.

Index Terms—Global Mobility, Machine Learning, Multi-Criteria Decision Support, Country Recommendation System, Gaussian Naïve Bayes, Explainable AI, Socio-Economic Indicators.

I. INTRODUCTION

Students and professionals seeking higher education, career advancement, and enhanced living conditions are now focused on global mobility. The major receiving countries, including the USA, Canada, UK, Germany, Australia, and Japan, have rapidly growing economies, safer living conditions, advanced healthcare systems and their education systems are widely respected throughout the world [1]–[3]. These factors make them one of the top ranked choices for global mobility [4]. It is important for the user to be aware of multiple criteria in their decision making process. When choosing among the top receiving countries, users must consider many different factors, such as living costs, employment opportunities, accepting migrants, levels of corruption, and the quality of living in these countries [5]–[7].

Traditionally, users have relied on social media to provide opinions of different countries, used personal judgment or made decisions based upon incomplete information, resulting in possibly inaccurate or biased decision making [8]. There is thus an increasing need for a user friendly, data-driven, and intelligent application that will objectively analyse countries and help users evaluate and select the best receiving country according to their needs [9].

The development of an ML- Multi-Criteria Evaluation System is to facilitate an intelligent choice of countries to study or work. An ML-Based Multi-Criteria Evaluation System will provide an objective and unbiased ranking for all countries based on global socio-economic indicators. By evaluating data using ML, we will generate a more accurate ranking of countries than traditional methods of decision support. Ultimately, we hope to empower people to make smart choices about their future through information that is representative of all aspects of their needs.

II. LITERATURE REVIEW

Over the past ten years, researchers have been utilizing increasingly sophisticated means of data-driven decision-making to identify optimal migration options and to evaluate countries [10]. The majority of the literature has focused on assessing only one of the various components of global mobility (e.g., opportunity for education) rather than developing a more comprehensive, multi-factor approach to creating opportunities to live, study or work in another country. An example of a traditional multi-criteria decision-making (MCDM) approach that has been used frequently to rank countries or study locations based on selected criteria (e.g., economic viability, safety and access to education) is the Analytic Hierarchy Process (AHP) [11], the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) [12] and the ELECTRE (method for ranking based on comparisons) [13]. These methods provide structure, however, they depend heavily on expert evaluation of the weighted criteria and the assignment of weights, both of which can lead to potentially biased assessments and inconsistencies in ranking decisions [14].

AI technologies are evolving rapidly, as a result, the researchers have utilized ML techniques to decrease the degree of Subjectivity when assessing countries, therefore providing greater reliability for country-level assessments [15], [16]. Researchers have utilized ML models such as Random Forests [17], XGBoost, SVMS, and Logistic Regression, which allow the researchers to identify hidden patterns, or non-linear patterns in the world's data, which is why these methods have begun to show promise in the field of immigration predictions, employment evaluations, immigration policies, as well as overall quality of life of countries [18]. The ability of ML is to automate decisions and improve the degree of accuracy of rankings from socio-economic data collected in the past.

There are still significant shortcomings associated with previous research efforts. Numerous studies investigate only a few principal factors (for instance: economic growth, security, education) and fail to provide an all-encompassing perspective of the global mobility decision-making process [19]. Further, several studies consider only specific geographic locations or government initiatives as part of their analysis of this topic. They do not include a broader consideration of the many different elements that impact an individual's choice to move abroad for study or work. On top of that, most of the models proposed by researchers overlook many of the major industrialized nations, which host millions of foreign students and professionals each year, and therefore do not take into account the preferences of these individuals when determining how to rank these countries [20]. As such, there are currently no full-fledged, multidimensional assessments of the outcomes of various countries based on multiple socio-economic factors that reflect individual migration preferences.

Using distinctive socio-economic indicators that typify developed economies, this research develops an intelligent decision-support solution by analysing six "advanced" nations, targeting Study or Work Outside the Home Country (WOSHC). Machine learning is employed to identify intricate patterns and correlations between these indicators. This approach to global mobility provides users with accurate, objective, and empirical results, rather than the subjective ranking methodology used in standard methodologies for multiple criteria decision making (MCDM).

III. METHODOLOGY

This paper includes a recommendation for an ML-based Multi-criteria Evaluation System to support decision making regarding global Mobility. The proposed pipeline methodology has been developed in a structured manner and involves several steps to create an end-to-end methodology of recommending the best Countries for a person looking to move globally. Figure 1 outlines the Workflow needed to develop the proposed ML-MCES system. The Workflow includes the steps of raw data preparation through to delivering a fully explained, recommended Country.

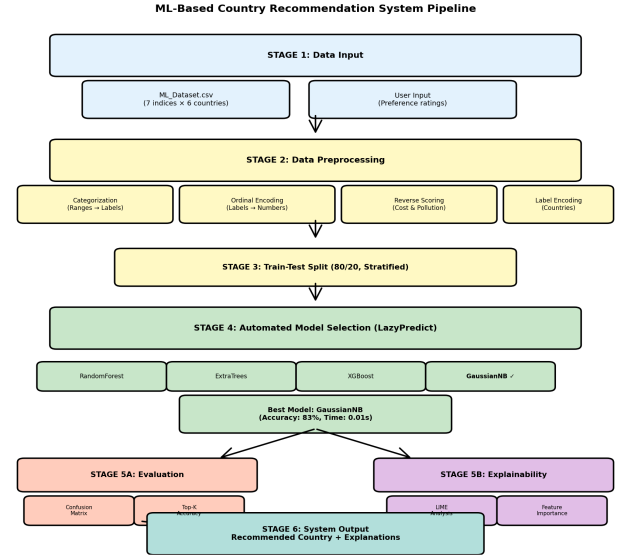


Fig. 1: Overall workflow of the proposed ML-MCES system.

A. Data Collection and Description

The socio-economic dataset used in this research is for 6 developed nations commonly evaluated for job and education relocation. Each country has 7 primary socio-economic measures:

- Quality of Life Index
- Purchasing Power Index
- Safety Index
- Health Care Index
- Cost of Living Index
- Pollution Index
- Education Index

All of these measures were collected from publicly available global data sources and saved in a structured CSV file format. This dataset's target variable is the country of residence (a multi-class label).

B. Feature Engineering and Categorization

For easier user interpretation of raw data, we have converted all indicators into categorical (ordered) levels: *Bad*, *Moderately Good*, *Good*, *Better*, and *Best*. This allows for more intuitive use of these numerical indices as well as better categorizing of numeric thresholds based on the specific characteristics of the domain. The conversion also aligns the order of indicators with user preference when using other indicators where low is preferable (Cost of Living, Pollution). This will enable the model to learn user relative preference, rather than just using absolute numbers.

C. Data Encoding

Ordinal Encoding was used to encode categorical features into numeric representations; this allowed for retaining the natural ordering of ratings and Social Costs; additionally, there were separate ordering schemes for benefit and cost-based indicators. The Country Designators were encoded numerically

using Label Encoding, enabling their utilization in a supervised learning environment.

D. Train-Test Split

The dataset was divided into two parts: Training and Testing using the 80/20 rule. To ensure equal distribution of each country's class in both training and testing datasets, Stratified Sampling was used. A fixed Random Seed was set to maintain reproducibility.

E. Model Selection Using LazyPredict

LazyPredict allowed for the evaluation of multiple classifiers through the use of supervised classification models. The performance of the classifiers was assessed through the use of Accuracy and Balanced Accuracy as metrics for determining the most reliable and consistent classifier with limited data - resulting in the selection of Gaussian Naïve Bayes, based upon its steady performance with the smallest number of observations.

F. Model Training and Evaluation

Using the trained model on the test dataset with encoded versions of the training dataset, model performance was evaluated by means of:

- Overall Accuracy
- Classification Report (Precision, Recall, F1-score)
- Confusion Matrix

In addition, Top K accuracy (Top-1, Top-3, Top-5) was computed to assess the quality of the recommendations. It is crucial to assess the quality of recommendations to determine if the correct country is found within the Top K recommendations for Decision Support Systems (DSS) rather than simply looking at correct classifications.

G. User-Based Recommendation System

The qualitative preferences of each indicator have been encoded within a user interaction module. The system will apply a default value for any criterion that the user does not specify, and will provide the user with an appropriate validation message when no value is entered for that criterion.

User Inputs are first preprocessed in the same way as the rest of the training data and then used in the trained Model to recommend a Country. In this way, users can receive recommendations based on their unique set of priorities.

H. Visualization and Comparative Analysis

To allow for better interactivity and richer presentation quality we produced various different types of visual representation, including:

- Confusion matrix heatmap
- Top-K accuracy bar chart
- Country recommendation comparison plots

These visual representations provide a comparative look across different user profiles as to how well they align in terms of their numerical metrics and visual trends.

I. Explainability Using LIME

To increase the level of transparency and trustworthiness we implemented Local Interpretable Model-Agnostic Explanations (LIME) methods to explain the rationale for each of the recommendations given to the users. LIME is a method that provides insight into which features contributed most heavily to the prediction, identifying items which helped or hurt the outcome.

The use of LIME on multiple simulated user profiles demonstrated how social/economic factors affect the choices for the countries recommended; thereby, confirming that the model's decisions are supported and can be reasonably interpreted through a common sense approach.

J. Methodological Summary

The methodology that is proposed on this occasion is that of an integrated system that consists of Automating the Selection of a Model, Ordinal Feature Engineering, Personalised Recommendation and the use of Explainable AI methods. With this methodology we are able to provide an excellent solution with respect to Accuracy, Fairness, Interpretability and Practical Application of the Global Mobility Decision Support System

IV. RESULTS AND DISCUSSIONS

In this section we present the experimental results obtained through the execution of ML-MCES (Machine Learning Based Multi Criteria Evaluation System). We provide a detailed analysis of the characteristics of the results produced through the ML-MCES through Multiple Classification Models, Recommendation Accuracy, Visualisation of Results and Explainability of the results.

A. Model Comparison and Selection

To compare the performance of classifiers for Country Recommendation, we evaluated several Multi-Class Classification Algorithms using LazyPredict. The dataset used contains Seven Socio-Economic Indicators as features and Six Target Countries (as Labels). LazyPredict automatically trained and evaluated 7 different Algorithms which are : Random Forest, Extra Trees, Gradient Boosting, K-Nearest Neighbors, Logistic Regression, XGBoost and Gaussian Naive Bayes. Of these, the Gaussian Naive Bayes (GaussianNB) Algorithm exhibited consistent and high performance even with a small sized dataset.

Performance Comparison of Machine Learning Classifiers (Top 10)

Model	Accuracy	Bal. Acc.	Time (s)
GaussianNB	0.833	0.833	0.019
ExtraTrees	0.833	0.833	0.213
ExtraTree	0.833	0.833	0.017
LabelPropagation	0.750	0.750	0.020
DecisionTree	0.750	0.750	0.022
Bagging	0.750	0.750	0.058
CalibratedCV	0.750	0.750	0.103
LogisticRegression	0.750	0.750	0.032
LinearSVC	0.750	0.750	0.029
LinearDiscriminantAnal...	0.750	0.750	0.042

Fig. 2: Performance comparison of machine learning classifiers evaluated using LazyPredict framework.

The selected model achieved:

- Accuracy: approximately 83%
- Balanced Accuracy: approximately 83%

GaussianNB's ability to maintain low levels of overfitting while providing stable generalization through a simple and probabilistic approach was an attractive feature for use with this dataset due to the small size, structure, and limitations inherent in working with such types of datasets.

B. Classification Performance

The chosen classifier was able to generate detailed performance metrics for each individual country within the sample test set when it was evaluated after training. Figure 3 shows the output confusion matrix for the selected model — GaussianNB.

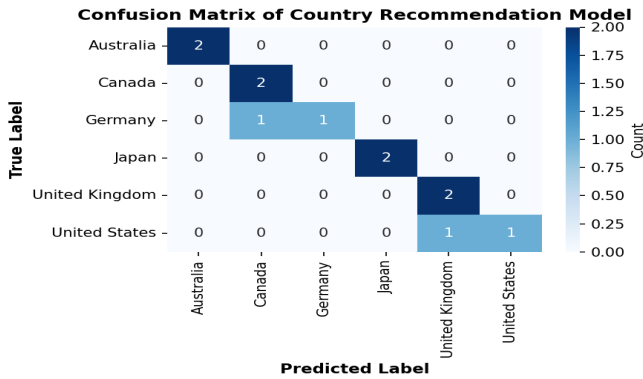


Fig. 3: Confusion matrix of the country recommendation model.

This model has been successful at classifying its target countries of Australia, Canada, Germany, and Japan as seen on the diagonal of the confusion matrix where there is a concentration of correct predictions; thus it is evident that there is a high degree of correct to incorrect predictions. As shown in Table 4, we can see the detailed precision (precision), recall (recall), and F1-scores (F1-score) for each of these country classes.

Classification Report - Country Recommendation Model

Country	Precision	Recall	F1-Score	Support
Australia	1.00	1.00	1.00	2
Canada	0.67	1.00	0.80	2
Germany	1.00	0.50	0.67	2
Japan	1.00	1.00	1.00	2
United Kingdom	0.67	1.00	0.80	2
United States	1.00	0.50	0.67	2
Accuracy			0.83	12
Macro Avg	0.89	0.83	0.82	12
Weighted Avg	0.89	0.83	0.82	12

Fig. 4: Classification report showing precision, recall, and F1-scores for each country class.

Australia, Canada, Germany, and Japan all received scores of 1.00 (perfect), and although the United Kingdom received precision scores of 0.67 and recall scores of 1.00 and the United States received precision scores of 1.00 and recall scores of 0.50, both of these countries were also performing competitively to the other countries listed. Finally, the overall, or macro-average, F1-score of the final model using this classifier was 0.82, indicating that there was a healthy balance across all countries' performances.

C. Final Model Evaluation

Following the selection of the optimum performing classifier, the retrained model was run against the test dataset, with the evaluation being based on performance metrics including, but not limited to: accuracy, precision, recall, and F1 score; and analysis of the confusion matrix.

The final test accuracy for the chosen model was 83.33 percent. The average macro F1 score was approximately 0.82, demonstrating that the classifier performs relatively evenly across all country classifications. The confusion matrix shows a majority of correct classifications are on the diagonal, or the strong diagonal dominance. That indicates that most samples were classified correctly; Small amounts of misclassifications occurred between countries with closely related socio-economic characteristics (e.g. Germany and Canada); it is reasonable to expect this phenomenon to occur given the small sample size and commonality among the indicator sets.

D. Top-K Accuracy Analysis

Since the proposed system is intended as a recommendation system rather than a strict classifier, Top-K accuracy was evaluated to measure recommendation quality.

The results are summarized as follows:

- Top-1 Accuracy: 0.83
- Top-3 Accuracy: 1.00
- Top-5 Accuracy: 1.00

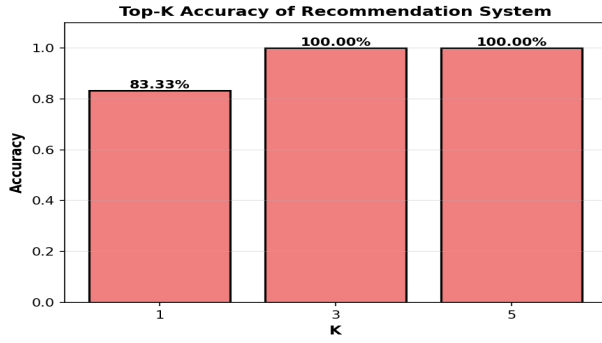


Fig. 5: Top-K accuracy of the recommendation system.

Based on these results, even if the first recommendation provided is incorrect, the actual country recommended would still appear in one of the top three recommendations. Therefore, these findings support the strength and dependability of the system for use in the real-world decision-making process.

E. User-Based Recommendation Analysis

The user personalization capability was validated by simulating five unique user types (profiles) with a variety of user priorities. Each of these profiles indicated which of seven user indicators were most important to them using a categorical preference scale.

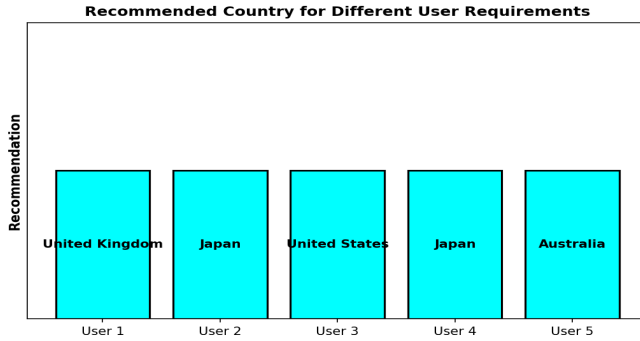


Fig. 6: Recommended countries for different user requirement profiles.

The recommended countries suggested by the model were based on the various user needs of these profiles and confirmed that the personalization system responds dynamically to user preference (rather than an inflexible/static outcome). This shows that the model's ability to provide country recommendations can accommodate a wide variety of mobility decision situations.

F. Visualization Consistency

A variety of visualizations were produced to support the evaluated numbers through:

- Confusion matrix heatmap
- Top-K accuracy bar chart
- User recommendation bar chart
- Requirement score heatmap

- Comparative requirement bar plots

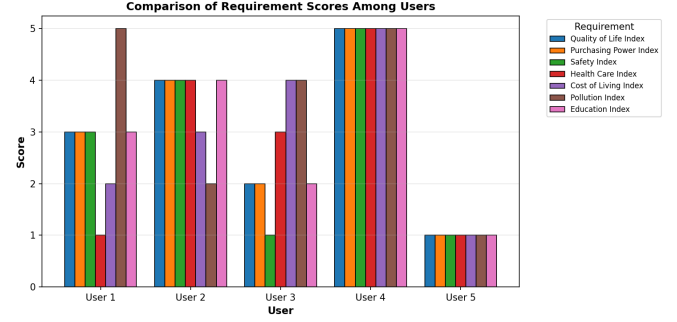


Fig. 7: Comparative requirement scores among users.

All visualisation representation supported the trends displayed in the quantitative evaluation metrics; demonstrating again that there is internal consistency within the Pipeline proposed.

G. Explainability Using LIME

To promote accountability, the authors have implemented a Local Interpretable Model-Agnostic Explanation Approach (LIME) to explain the output of each model. Using LIME on the Environmental and Social Science Profiles of each user, the authors have identified the following features contributing positively and negatively to the recommended country for the user: Quality of Life, Safety, Cost of Living, Education.

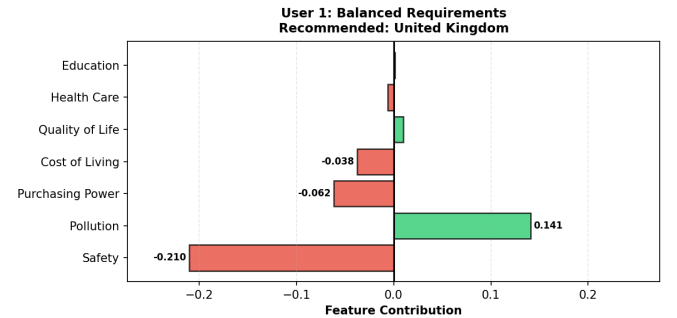


Fig. 8: LIME explanation for User 1: Balanced Requirements.

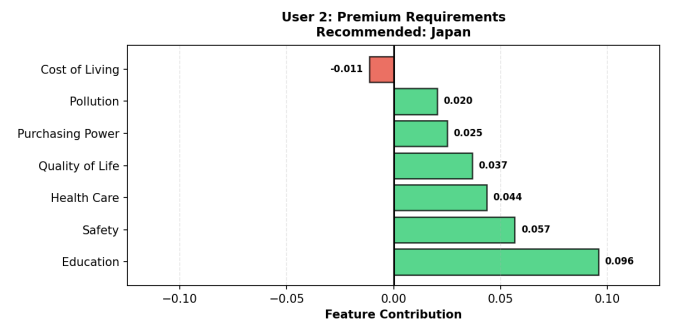


Fig. 9: LIME explanation for User 2: Premium Requirements.

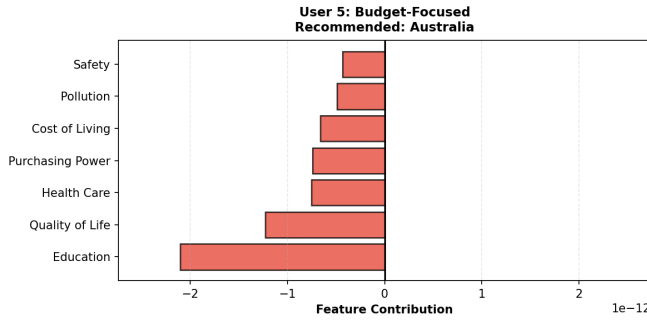


Fig. 10: LIME explanation for User 5: Budget-Focused.

The green bars on the LIME visualisation represent positive contributions; while the red bars indicate negative influences. The explanations are consistent and interpretable, demonstrating that the recommendations were based upon logical economic and social factors and not random opportunity.

H. Discussion

The proposed ML-MCES framework has been shown to have reasonable predictive accuracy, generate credible recommendations, and offer strong interpretability on a limited amount of data. The consistency observed between the evaluation metrics, graphical representations and explainability of the recommendation process indicate the robustness of the ML-MCES framework. Furthermore, the framework's ability to provide personalised, accurate Top-K recommendations along with relevant explanations demonstrates the framework's applicability to real time applications such as Global Mobility Decision Support Systems.

I. Limitations

While the system has yielded encouraging results, it is important to note that there are important limitations. First, the dataset contains information only from six developed nations; therefore, any generalised findings from this data cannot be applied to the developing regions. Second, the way that these socio-economic indicators were collected means that all are collected on a national basis with no regard to potential regional variations that may exist within nations. Third, converting continuous indicators into categorical levels makes it easier to interpret, but this process makes it harder for the model to detect small differences in socio-economic conditions. Fourth, although Gaia's Naïve Bayes method provided good results, the assumptions regarding the independence of one feature to another may not accurately represent the relationships and dependencies between several factors.

V. CONCLUSION AND FUTURE WORK

The research discusses an ML based Multi-Criteria Evaluation System (ML-MCES) for supporting Global Mobility decision-making. It includes socio-economic indicators, software for automated model selection, and an Explainable Artificial Intelligence component for enhanced transparent

decision-making. Tests using Gaussian Naïve Bayes achieved 83.33% accuracy.

Future research will include expanding the dataset, developing additional criteria, utilizing real-time data, and increasing scalability through the use of ensemble modeling and interactive user interfaces.

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